

Hemoglobinopathies

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Outlines

- ❑ **Structure of human hemoglobin**
- ❑ **Disorders of hemoglobin:**
 - ✓ **Structural globin chain variants:**
 - Sickle cell anemia**
 - ✓ **Disorders of synthesis of the globin chain:**
 - **Thalassemia**
 - **HPFH**

Hemoglobin (Hb)

Hb is found in RBCs, its main function is to *transport O₂ to tissues*.

Structure: 2 parts : heme + globin

Globin: four chains.

Heme: Porphyrin ring with central iron. Iron is the site of attachment with O₂.

Hemoglobin (Hb)

There are 4 heme groups, each attached to on globin chain.

So one Hb molecule can carry up to 4 O₂ molecules.

According to sequence of amino acids in the primary structure of each globin chain, there are four types of chains; α , β , γ and δ .

Human hemoglobins

Hb

Structure

%In adult

Gower I

$\xi_2 \epsilon_2$ (zeta 2 epsilon 2)

Embryonic

Gower II

$\alpha_2 \epsilon_2$ (alpha 2 epsilon 2)

Portland I

$\xi_2 \gamma_2$ (zeta 2 gamma 2)

Fetal

HbF

$\alpha_2 \gamma_2$ (alpha 2 gamma 2)

<1%

HbA

$\alpha_2 \beta_2$ (alpha 2 beta 2)

97-98%

Adult

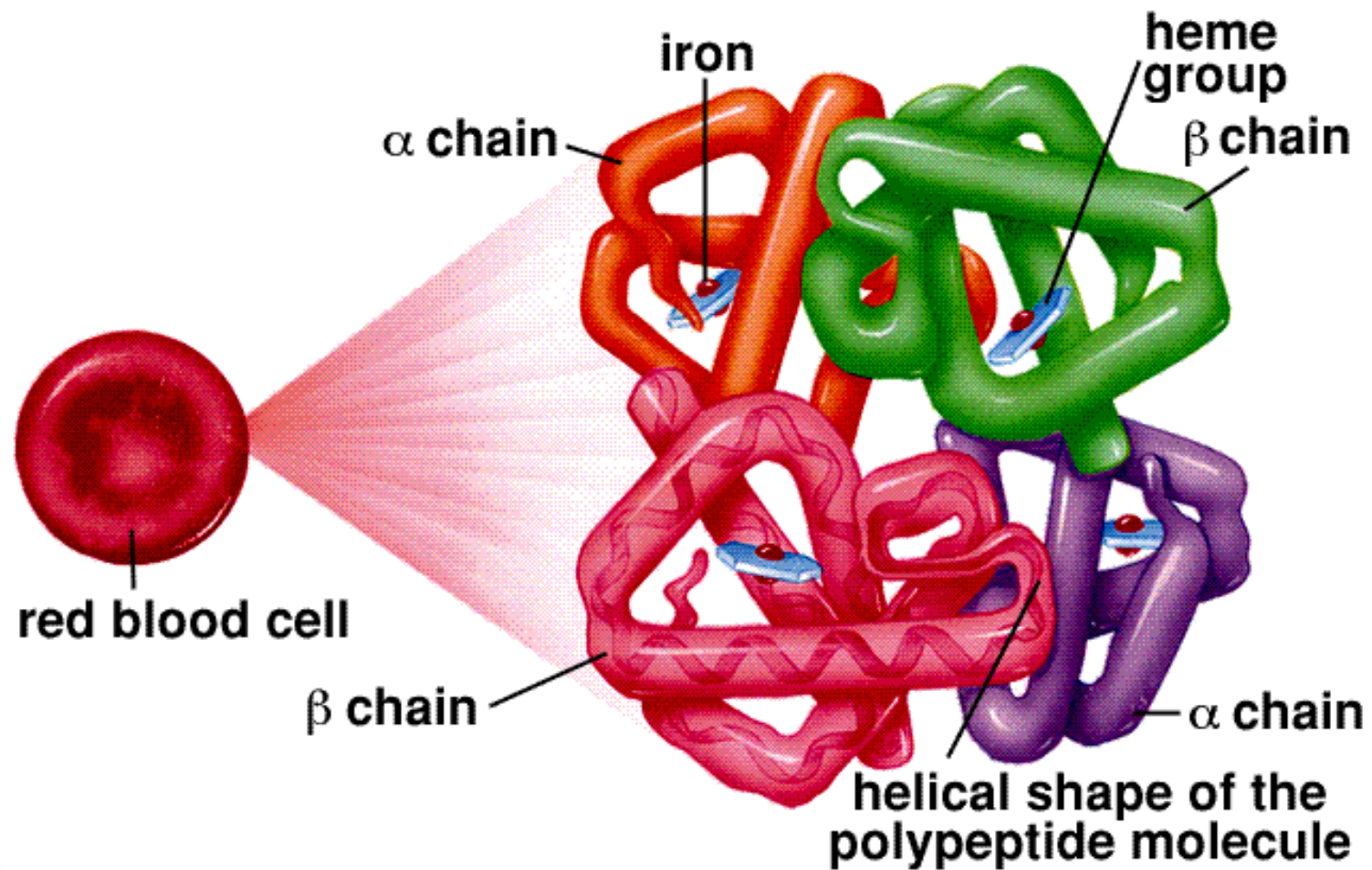
HbA2

$\alpha_2 \delta_2$ (alpha 2 delta 2)

2-3 %

- Adult hemoglobin is a $[\alpha(2):\beta(2)]$ tetrameric hemeprotein found in erythrocytes .
- It is responsible for *binding oxygen in the lung* and transporting the bound oxygen throughout the body where it is used in aerobic metabolic pathways.
- The protein portion of each of these chains is called "*globin*". The α and β globin chains are very similar in structure
- Each *alpha chain* contains *141* amino acids , while the *beta chain*, contains *146* amino acids.

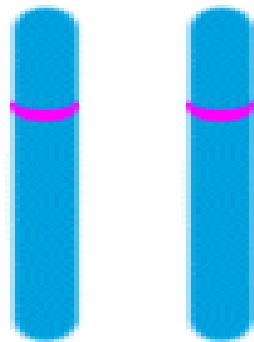
Hemoglobin Molecule



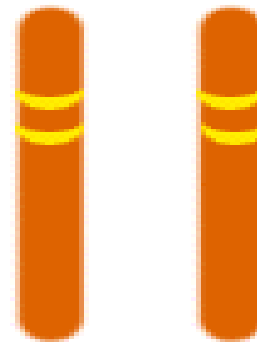
Hemoglobin synthesis

- Hemoglobin (Hb) is synthesized in a complex series of steps.
- The **heme part** is synthesized in a series of steps in the mitochondria and the cytosol of immature red blood cells.
- The globin protein parts are synthesized by ribosomes in the cytosol.
- Production of Hb continues in the cell throughout its early development from the proerythroblast to the reticulocyte in the bone marrow.

Chromosomes

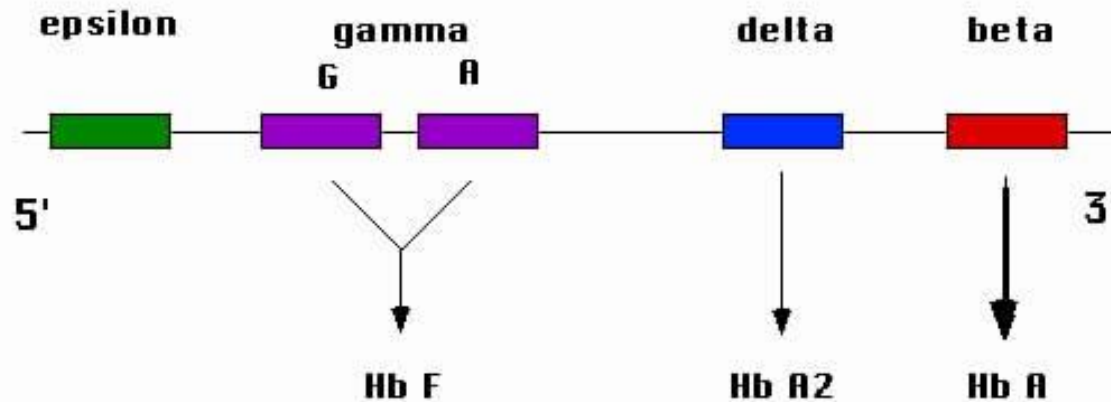


Chromosome 11
 β globin gene

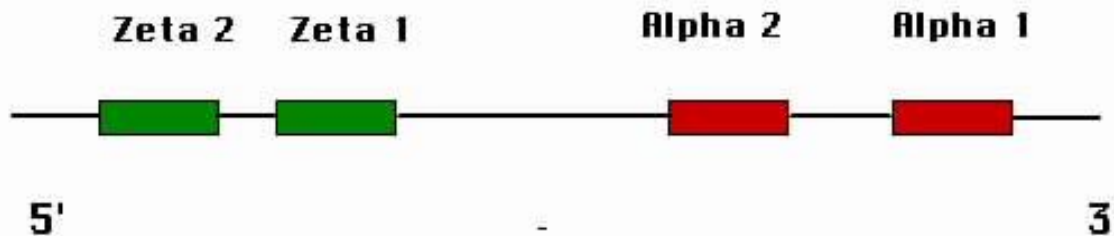


Chromosome 16
 α globin gene

Beta Globin Gene Cluster Chromosome 11



Alpha Globin Gene Cluster Chromosome 16



Disorders of hemoglobin

Structural globin chain variants: qualitative defects that results from single amino acid substitution in alpha or beta chains of adult Hb.

e.g. Sickle –cell disease.

Disorders of ***synthesis of the globin chains***; quantitative defects.

e.g. The thalassemia

Sickle cell Hemoglobinopathy

- Sickle cell anemia (disease) (Hb SS)
- Sickle cell trait (Hb AS)

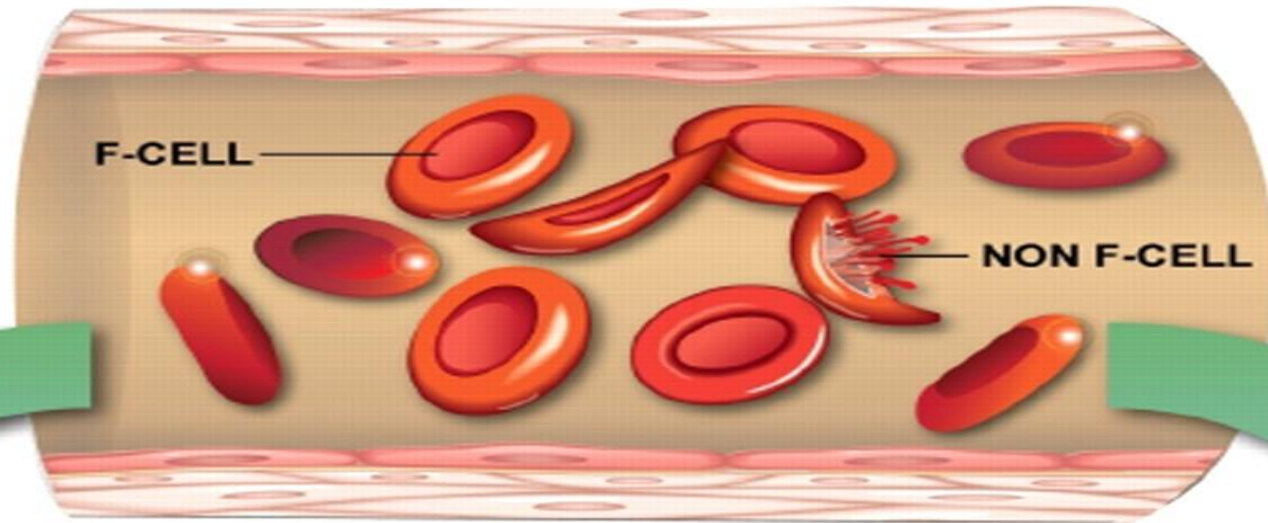
Sickle-cell disease

Autosomal recessive

Hb –S

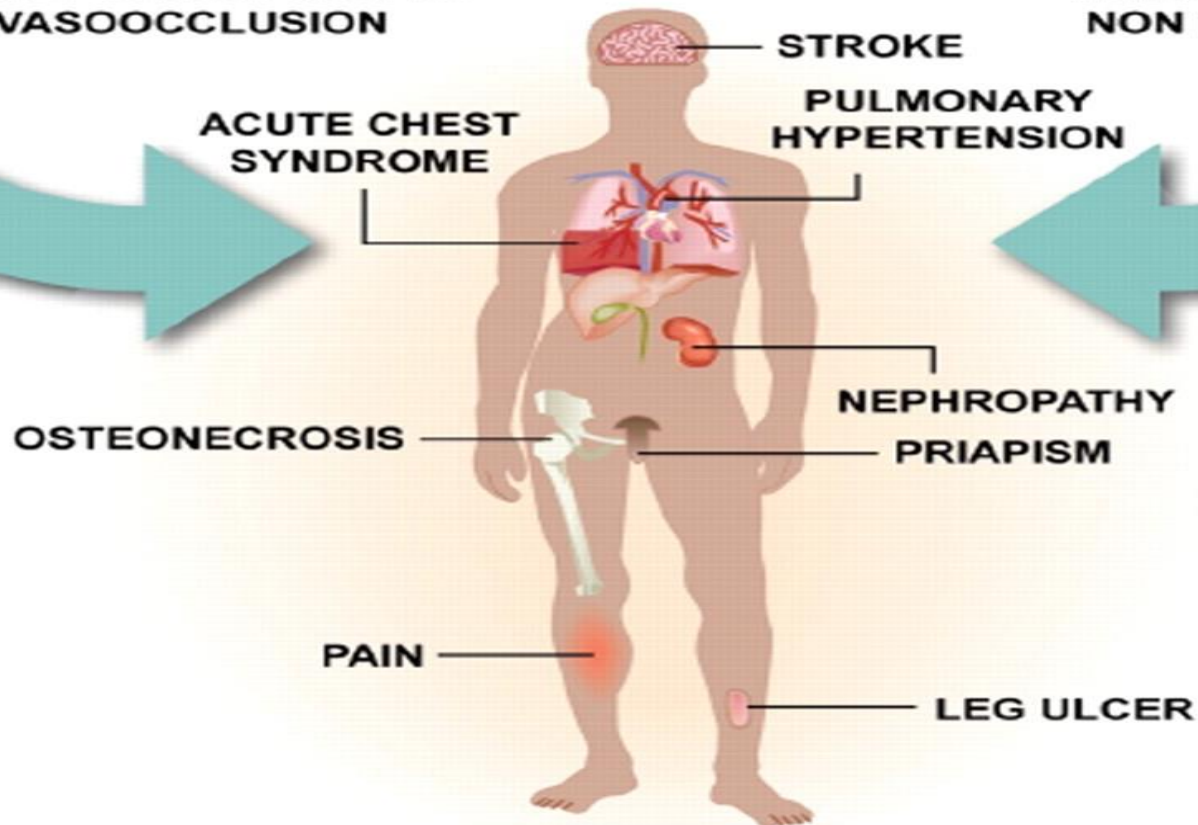
A genetic life-long blood disorder characterized by red blood cells that assume an abnormal, rigid, sickle shape.

Sickling decreases the cells' flexibility and results in a risk of various complications. The sickling occurs because of a mutation in the globin gene.



**MANY F-CELLS PREVENT
VASOOCCLUSION**

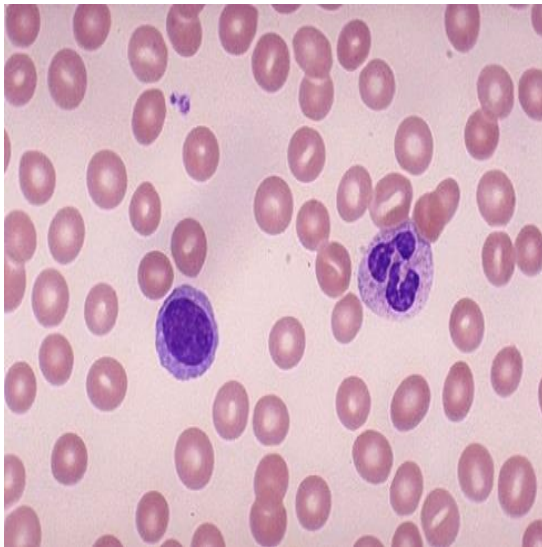
**HEMOLYSIS OF
NON F-CELLS**



Normal Vs. Sickle Red Cells

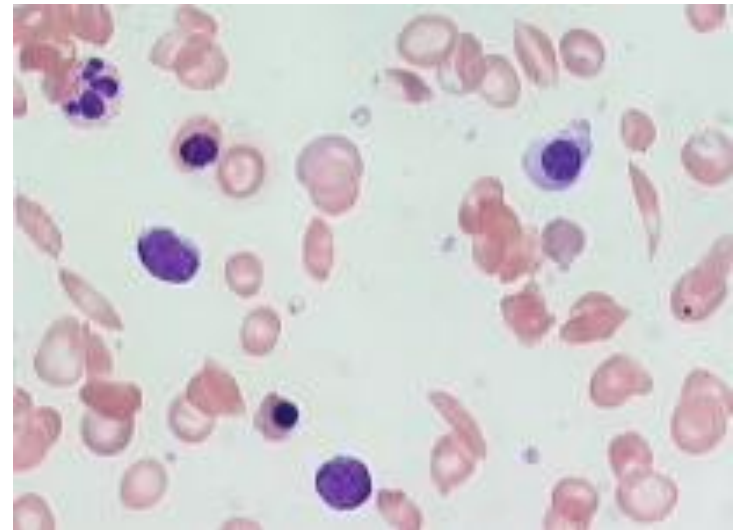
Normal

- Disc-Shaped
- Deformable
- Life span of 120 days



Sickle

- Sickle-Shaped
- Rigid
- Lives for 20 days or less



Sickle cell anemia

AR

- Sickle cell disease, is caused by a point mutation in the β globin gene.
- As a result of this mutation, valine (a non-polar amino acid) is inserted into the β globin chain instead of glutamic acid (an electrically charged amino acid). (HbS)
- The mutation causes the RBCs to become **stiff** and **sickle-shaped** when they release their load of oxygen.

Sickle cell anemia

AR

- The sickle cell mutation produces a "sticky" patch on the surface of the β chains when they are not complexed with oxygen.
- Because other molecules of sickle cell hemoglobin also develop the sticky patch, they adhere to each other and polymerize into long fibers that distort the RBC into a sickle shape.
- Life span of 20 days (normal RBCs life span 120)

Symptoms:

- Anemia.
- Jaundice.
- Delayed growth
- Episodes of pain
- Hand-foot syndrome
- Frequent infections.
- Vision problems.
- Sickle Cell Crisis

- **Sickle Cell Crisis**

- Episodes of **pain** that occur with varying frequency and **severity** in different patients and are usually followed by periods of remission.
- Severe sickle cell pain has been described as being equivalent to **cancer pain** and more severe than post surgical pain.
- It most commonly occurs in the **lower back, leg, abdomen, and chest**, usually in two or more locations. Episodes usually recur in the same areas.

Sickle trait

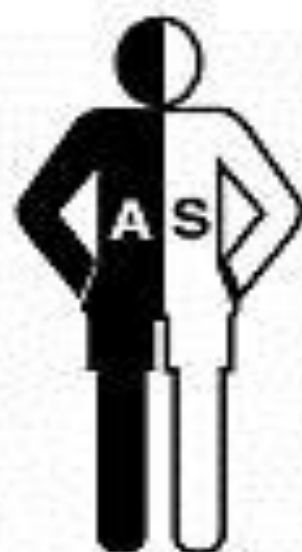
Hb AS , heterozygous

- People who inherit just a single gene are referred to as having the *sickle trait*.
- These people are protected **against malaria** and do not develop sickle cell disease.
(Heterozygous advantage)

Sickle trait

Risk factors:

- sudden death during vigorous exercise.
- Vaso occlusive manifestation from sever hypoxia(high altitude, flying in unpressurized air craft)
- About 40% of people in certain parts of Africa and about 9% of African-Americans have the trait.




normal hemoglobin
25% chance


sickle cell trait
50% chance


sickle cell trait
50% chance


sickle cell disease
25% chance



- **Investigations**

CBC: anemia

blood film: sickle cells

Hb electrophoresis: HbS

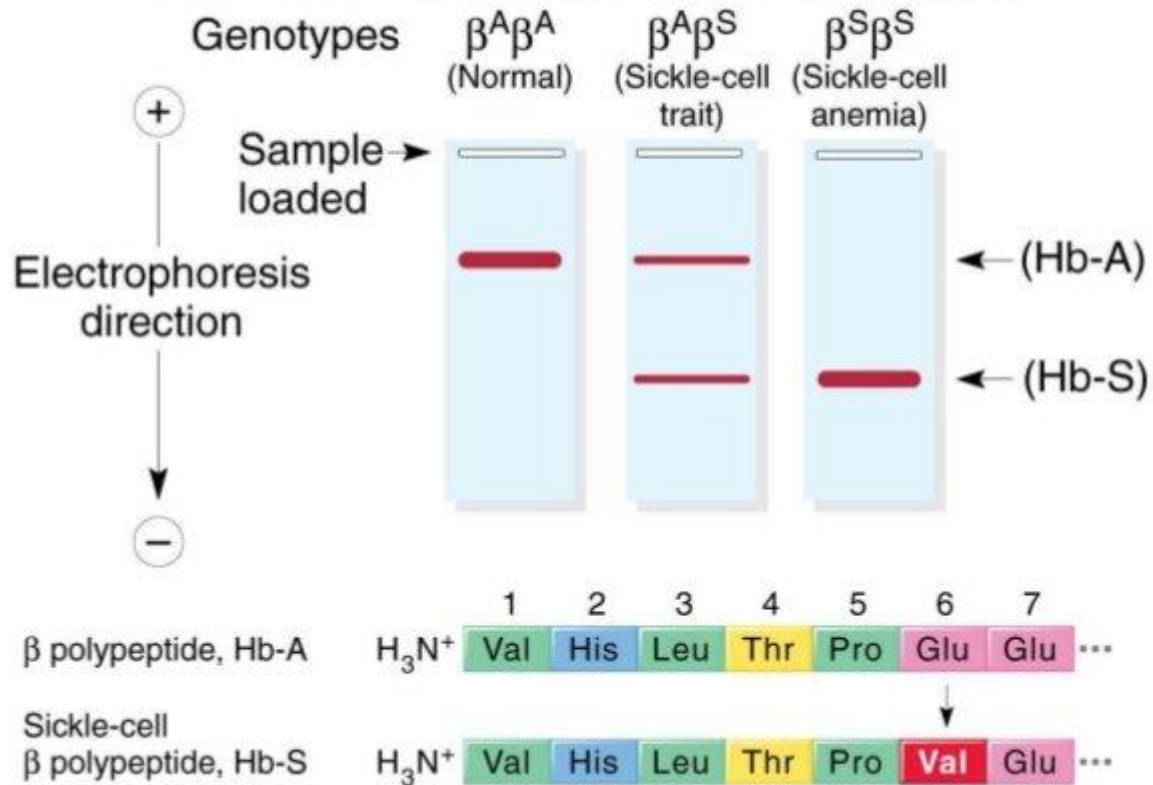
Genetic testing to detect gene variant

- **Treatment:**

Prophylactic penicillin

Hydroxy urea: to increase HbF

Electrophoretic pattern of Sickle Cell Disease



PJ Russell, *iGenetics 3rd ed*

Disorders of hemoglobin synthesis

- *Thalassemias*

Quantitative defect characterized by reduce rate of one or more of the globin chains

- *Hereditary persistence of high fetal hemoglobin (HPFH)*

Thalassemia

- Thalassemias are the commonest single group of inherited disorders in humans, occurring in persons from **mediterranean area, the middle east, the Indian subcontinent and South-East Asia.**
- They are a heterogeneous group of disorders and are **classified according to the particular globin chain or chains synthesized in reduced amounts, e.g. α , β , δ β -thalassemia.**

Thalassemia

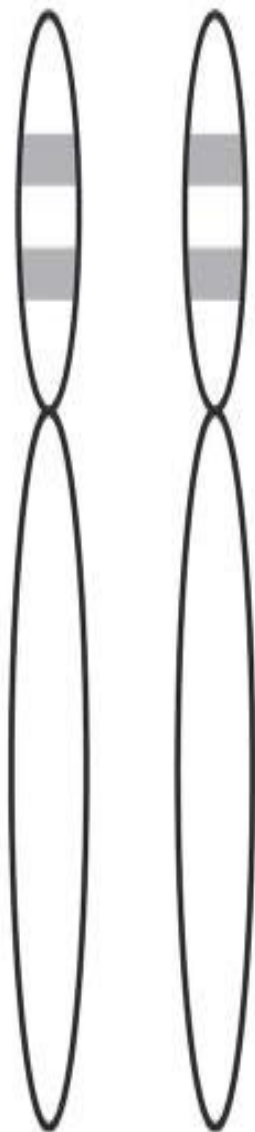
- The pathophysiology of thalassemia:

An imbalance of globin-chain production results in the accumulation of free globin chains in the red blood cell precursors, which, being insoluble, precipitate, resulting in hemolysis of the red blood cells, i.e. hemolytic anemia, with consequent compensatory hyperplasia of the bone marrow

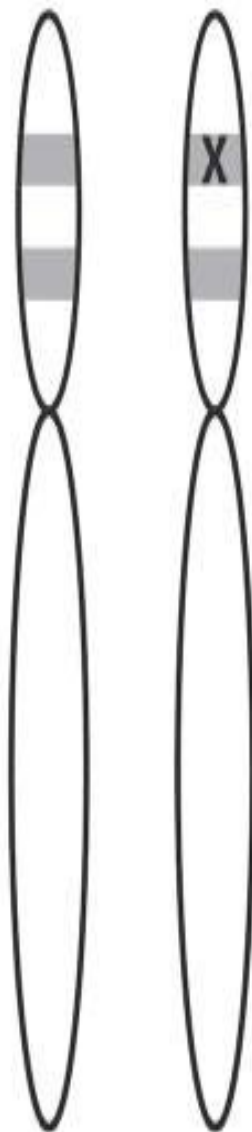
α -thalassemia

- Hemoglobin Bart hydrops fetalis (Hb Bart) syndrome
- Hemoglobin H (HbH) disease

Normal



Alpha-globin
mutation



Key

Chromosome 16



Functional alpha-
globin allele



Nonfunctional
alpha-globin allele

Genetic basis of Alpha Thalassemia

- Encoding genes on chromosome 16 (short arm)
- Each cell has 4 copies of the alpha globin gene
 - Each gene responsible for $\frac{1}{4}$ production of alpha globin
- 4 possible mutation states:
 - Loss of **ONE** gene → **silent carrier** (Alpha heterozygote)
 - Loss of **TWO** genes → thalassemia **minor** (trait)(Alpha heterozygote)
 - Loss of **THREE** genes → **Hemoglobin H**
 - Accumulation of beta chains
 - Association of beta chains in groups of 4 → Hemoglobin H
 - Loss of **FOUR** genes → Hemoglobin **Barts**
 - NO alpha chains produced ∴ only gamma chains present
 - Association of 4 gamma chains → Hemoglobin Barts

1-Hemoglobin Bart (Hb Bart)

Hydrops fetalis syndrome

- α^0 thalassemia (γ_4)
- All four α -globin alleles are deleted or dysfunctional (inactivated).
- Analysis of Hb present in fetuses with hydrops fetalis reveals it to be formed of four γ -globin chains (Hb Barts)
- The most severe form of α -thalassemia, is characterized by:
 - fetal onset of generalized edema, ascites, pleural and pericardial effusions, and severe hypochromic anemia.

1-Hemoglobin Bart (Hb Bart)

Hydrops fetalis syndrome

- α^0 thalassemia (γ_4)

Prenatal diagnosis:

By ultrasonography at 22 to 28 weeks' gestation and can be suspected in an at-risk pregnancy at 13 to 14 weeks' gestation when increased nuchal thickness, possible placental thickness, and increased cardiothoracic ratio are present.

- Death in the neonatal period is almost inevitable.

Hydrops fetalis syndrome

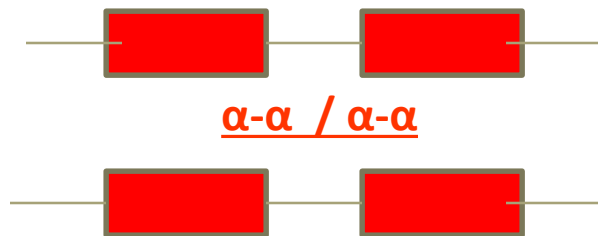
- **Clinical picture**
- **Sever anemia leads to congestive heart failure>>>>> generalized edema.**
- **Cause of death: heart failure**
- **Tissue hypoxia (O₂ dissociation curve shifted to the left)**

2-Hemoglobin H (HbH) disease

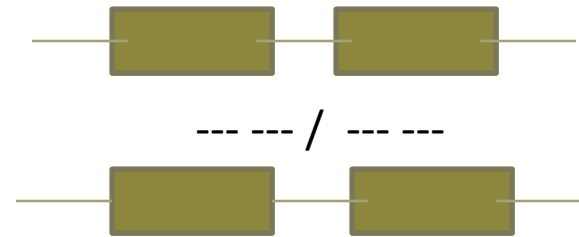
- Hb H: (β_4) > 25%
- should be suspected in an infant or child with **a mild-to-moderate (rarely severe) microcytic hypochromic hemolytic anemia and hepatosplenomegaly**. Scleral icterus, cholelithiasis.
- **Mild thalassemia-like bone changes** are present in approximately one-third of affected individuals.
- Unlike Hb Bart syndrome, HbH disease is **compatible with survival** into adulthood.
- HbH disease is a result of deletion or dysfunction of three of four α -globin alleles.

Mutational basis of **Hydrops Fetalis**

All four α -globin alleles are deleted or dysfunctional (inactivated)



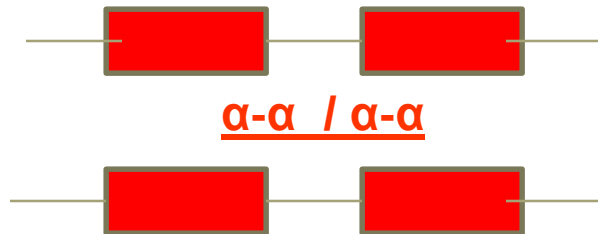
Normal
 $\alpha\text{-}\alpha / \alpha\text{-}\alpha$



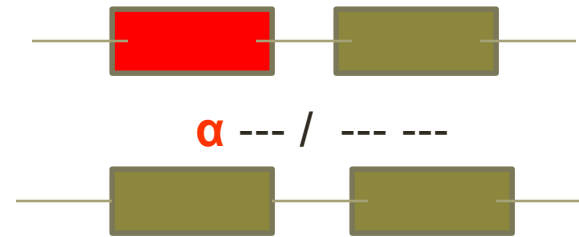
Hydrops Fetalis
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Mutational basis of Hb H disease

HbH disease is a result of deletion or dysfunction of three of four α -globin alleles.



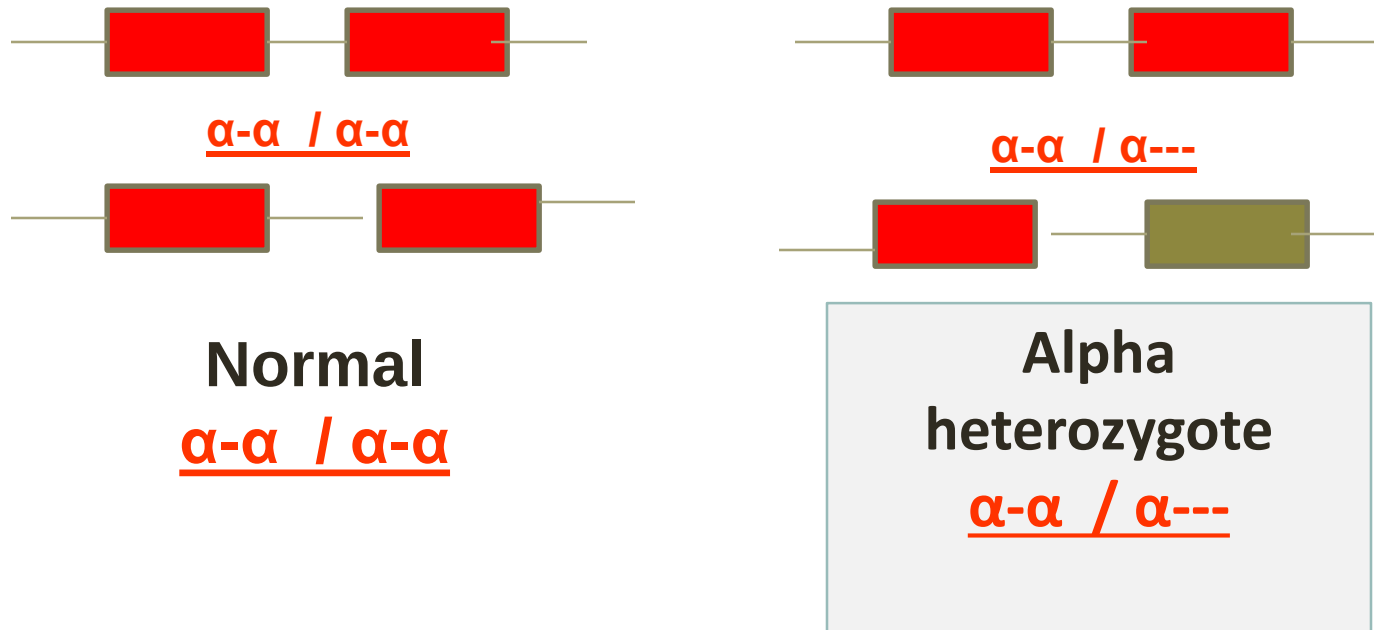
Normal
 $\alpha\text{-}\alpha / \alpha\text{-}\alpha$



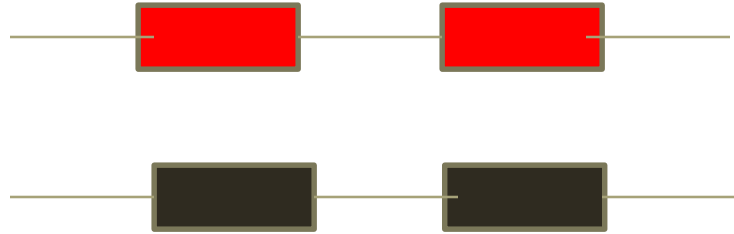
Hb H disease
 $\alpha \text{ --- } / \text{ --- --- }$

Mutational basis of α -globin alleles

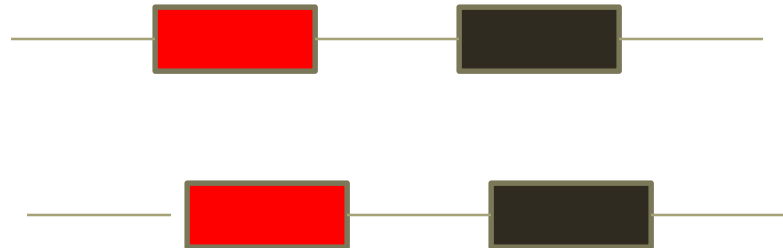
Alpha-thalassemia also has two heterozygote (carrier) states



α - α / α --- **thalassemia** usually results from deletion or dysfunction of one α -globin gene.



α - α /-- -- , α -- / α - thalassemia
generally results from deletion or dysfunction of
two α -globin genes.



Alpha heterozygote

β -thalassemia

- β -thalassemia is caused by underproduction of the β -globin chain of hemoglobin.
- **Different types of mutations show:**
 - reduced β -globin chains (β^+)
 - absent β -globin chains (β^0)
- Persons homozygous for β^0 –thalassemia mutations have a severe transfusion-dependant anemia.

Mutational basis of β -thalassemia

- The various types of mutation are often unique to certain population groups.

six main functional types

1-Transcription mutations

2-mRNA splicing mutations

3- Polyadenylation signal mutations

4-RNA modification mutations

5-Chain termination mutations

6- Missense mutations

Mutational basis of β -thalassemia

1-Transcription mutations

- **Site:** 5' flanking TATA box or promoter region
- **Results in :** reduced transcriptional level of the β -globin gene

2-mRNA splicing mutations

- **The most common mutation in Mediterranean region**
- **Site:** splicing sites 5'GT or 3'AG of the introns
- **Results in:** reduced levels of β -globin mRNA.

- **3- Polyadenylation signal mutations**
- **Site:** 3' end of the untranslated region of β -globin gene
- **Results in:** loss of signal of cleavage and polyadenylation of the β -globin gene transcript
- **4-RNA modification mutations**
- **Site:** 5' and 3' DNA involved in capping and polyadenylation of mRNA
- **Results in:** abnormal transportation of β -globin mRNA to cytoplasm + reduced levels of translation.
- **5-Chain termination mutations**
- **Site:** nonsense chain termination codon
- **Results in:** premature termination of translation of the β -globin mRNA
- **6- missense mutations**
- **Lead to highly unstable β -globin**

Clinical aspects of β -thalassemia

1-Thalassemia major or Cooley,s anemia (β^0)

2- Thalassemia trait or thalassemia minor (β^+)

3- δ β -thalassemia

1-Thalassemia major or Cooley's anemia:

- present in the first year after age 6 months???? why ?????
- Increase Alpha chains, are unstable lead to formation of inclusion bodies precipitates on the inner surface of red cell membrane leading to damage and hemolysis.
- With a severe transfusion-dependant anemia.
- Compensatory expansion of the bone marrow results in an unusually shaped face and skull.

2- Thalassemia trait or thalassemia minor

Individuals heterozygous for β -thalassemia, usually have no symptoms or signs. They have microcytic, hypochromic anemia, which can often be confused with iron deficiency anemia.

δ β -thalassemia:

- There is underproduction of both the δ and β - globin chains.
- δ β -thalassemia has been shown to be caused by extensive deletions in the β -globin region involving the δ and β - globin structural genes.

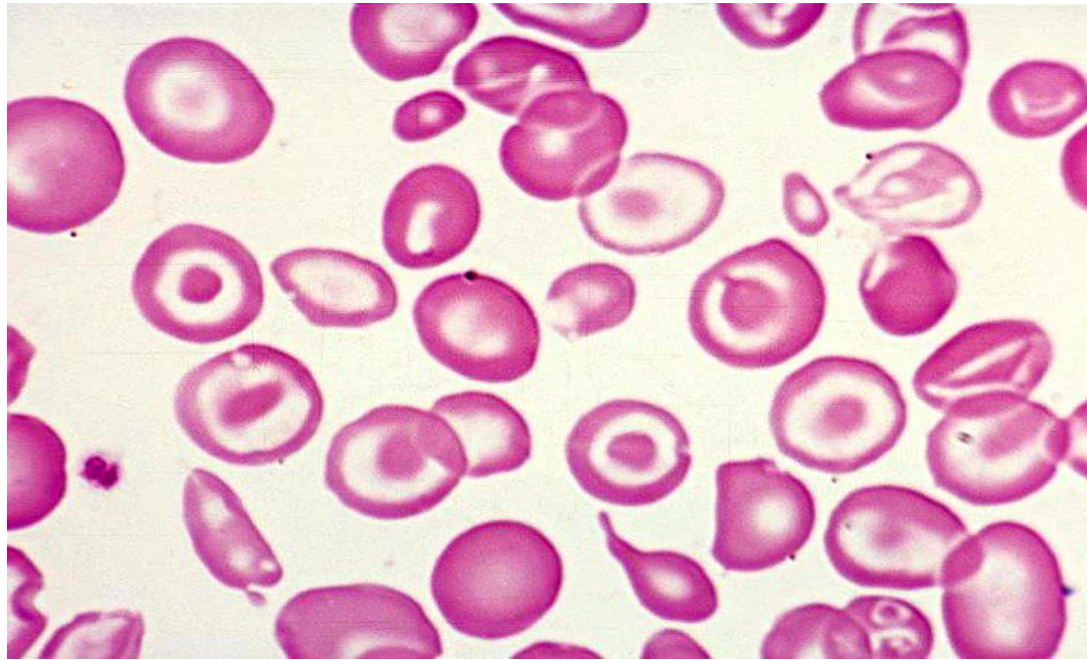
Diagnosis of thalassemia

Laboratory results

In thalassemia major,

- Show lowered RBC count and Hb level, microcytosis, elevated reticulocyte,
- Bilirubin, urinary and fecal urobilinogen levels.
- A peripheral blood smear reveals target cells, microcytes, pale nucleated RBCs, and marked anisocytosis.

Diagnosis of thalassemia



Diagnosis of thalassemia

- X-rays of the skull and long bones show thinning and widening of the marrow space because of overactive bone marrow.
- The bones of the skull and **vertebrae may appear granular**; **long bones** may show areas of **osteoporosis**. The **phalanges** may also be **deformed** (rectangular or biconvex).
- Quantitative Hb studies show : (Hb electrophoresis)
 - Rise in HbF
 - A slight increase in HbA₂
- Molecular to detect the type of mutation.



X-rays of the skull show thinning and widening of the marrow space because of overactive bone marrow

Treatment of Thalassemia

Thalassemia trait

Normally, there are no treatments recommended.

Thalassemia major

- Regular **blood transfusions**, usually every four weeks.
- Chelating agent **(desferrioxamine)**
- **Splenectomy** (removal of the spleen).
- **Bone marrow transplants**
- New treatment: Gene therapy, Stem cell therapy

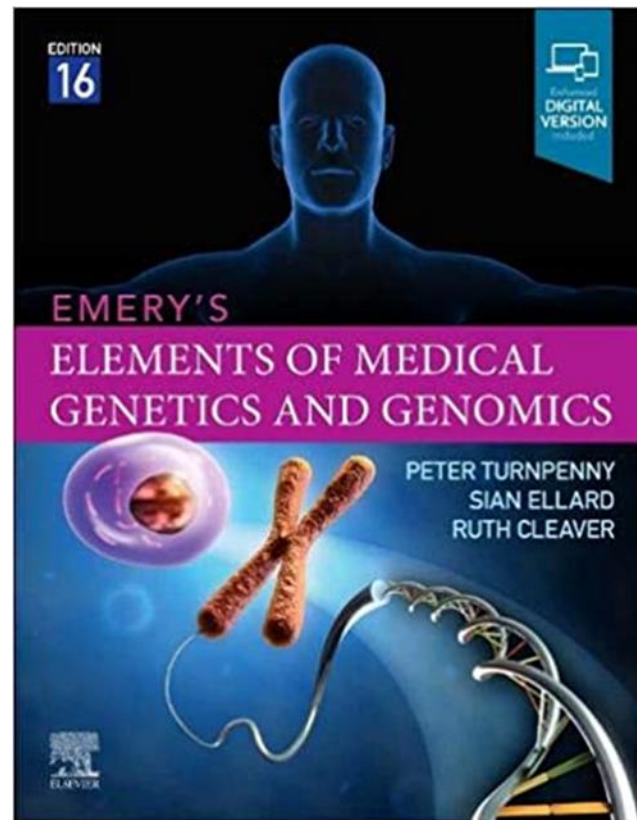
Hereditary persistence of fetal hemoglobin

- $\alpha_2 \gamma_2$ (alpha 2 gamma 2)
- Hereditary persistence of fetal hemoglobin (HPFH), is a benign condition in which significant fetal hemoglobin (hemoglobin F) production continues well into adulthood.
- The condition is usually asymptomatic, and is only noticed when screening for other hemoglobin disorders.

Hereditary persistence of fetal hemoglobin

Types of mutations:

- deletions of the δ and β - globin genes.
- Point mutations, in the promoter region of one or another gamma-globin gene, that are thought to alter interactions between various transcription factors and the promoter.



Emery's Elements of Medical Genetics 17th edition 2022 by Turnpenny and Ellard.

(Chapter 12, Hemoglobin and the Hemoglobinopathies, p 161)